

**SEDAR 77 HMS Hammerhead Sharks****1. STOCK ID***Genetics Working Group*

Smooth Hammerhead Sharks - We caution that no sampling or analyses included U.S. Caribbean jurisdictions (Puerto Rico and U.S. Virgin Islands) and recommend sampling efforts to determine if Smooth Hammerheads occur in the U.S. Caribbean jurisdictions and, if so, determine whether or not they are genetically differentiated from the core U.S. Atlantic population

Great Hammerhead Sharks - We recommend sampling and genetic analyses from the U.S. Caribbean jurisdictions. We also recommend sampling and genetic analysis of young-of-the year and small juvenile (< 110 cm total length) individuals because the current sample is dominated by individuals in the mobile phase of their life-cycle, which could mask structure based on reproductive philopatry.

Carolina Hammerhead Sharks - We caution that no sampling or analyses included U.S. Caribbean jurisdictions. We recommend sampling efforts to determine if Carolina Hammerheads occur in the U.S. Caribbean jurisdictions and, if so, determine whether or not they are genetically differentiated from the core U.S. Atlantic population.

Scalloped Hammerhead Sharks - We recommend genetic analyses of U.S. Caribbean Scalloped Hammerheads as a matter of urgency given that the Central & Southwest Atlantic Distinct Population Segment (DPS) of this species is listed under the Endangered Species Act (<https://www.federalregister.gov/documents/2014/07/03/2014-15710/endangered-and-threatened-wildlife-and-plants-threatened-and-endangered-status-for-distinct>).

*Spatial Movements and Catches Working Group*

Overall, the movement/migration data available for great hammerheads from the Gulf of Mexico and Atlantic Ocean is limited and the Spatial-Movement Workgroup recommends additional tagging (conventional, acoustic, and satellite) studies to better elucidate their movement patterns within the region.

Overall, spatial data, especially with respect to movements and migrations, available for smooth hammerhead from the Gulf of Mexico and Atlantic Ocean is scant at best and the Spatial-Movement workgroup recommends additional tagging (conventional, acoustic, and satellite) studies to better elucidate their movement patterns within the region.

Overall, the movement/migration data available for scalloped hammerhead from the Gulf of Mexico and Atlantic Ocean is limited and the Spatial-Movement workgroup recommends additional tagging (conventional, acoustic, and satellite) studies to better elucidate their movement patterns within the region.

Overall, the movement/migration data available for Carolina hammerhead from the Atlantic Ocean is non-existent and the Spatial-Movement workgroup recommends tagging (conventional, acoustic, and satellite) studies to better elucidate their movement patterns within the region.

A more in depth analysis than was possible for this stock ID workshop could be undertaken in the future as a research topic to help differentiate between stocks using catch and effort data and considering the effect of mis-identification, especially for recreational fisheries.

#### *Catches Working Group*

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## **2. DATA WORK SHOP RECOMMENDATIONS**

### *Life History*

- Increase data and sample collection in all forms necessary for informing age related parameters for all hammerhead species, with particular attention to Carolina and smooth hammerheads of both sexes and female scalloped hammerheads.
- Investigate alternative methods for non-lethal estimation of age and/or maturity status (e.g., epigenetic ageing). Conduct age validation studies on scalloped hammerheads to reduce uncertainty in band counting methodology.
- Increased reproductive sampling for all species throughout their range, especially with regard to brood size, gestation period, and reproductive cycle.
- Improve standardization of reproductive measurements and sampling techniques across research groups to facilitate better estimates of reproductive parameters.
- Increase genetic surveillance of scalloped and Carolina hammerheads in the Atlantic in order to further delineate species-specific life history traits and important habitats
- Continued genetic monitoring of Carolina and scalloped hammerheads within nurseries to track the relative abundance of the two species.
- Determine life-stage specific movement patterns and habitat utilization for all hammerhead species using electronic tagging, with particular attention to identifying pupping areas for great and smooth hammerheads.
- Assess stock structure and movement between Caribbean and U.S. waters for scalloped and great hammerheads.
- Identify species-specific abiotic characteristics driving distributions and how environmental changes could impact the life history and distribution of hammerheads in the western North Atlantic Ocean.

### *Catches*

- Increase public education outreach activities for species identification in the recreational fishery. This is important because there are no species identification training workshops for recreational fishers, and it is difficult to distinguish among different species, especially juveniles, by non-trained individuals.
- Improve the MRIP process to filter biased sampling that leads to unreal, extreme fluctuations in catch data for sharks, through a QA step that is applied with an objective, non-arbitrary procedure.
- Promote that the next stock assessment of hammerhead shark species/stocks be conducted under the auspices of an RFMO (e.g., WECAFC) so that all sources of removals and abundance indices and length compositions (if available) from Caribbean nations where the species/stock is distributed can be accounted for.
- Pooling observed sets for all areas by either each observed year or all observed years without considering variance of areas and seasons, along with an assumption of effort (number of logbook hooks) being a known constant, may cause the actual variance of discard estimates to be underestimated. This in turn will produce a narrower confidence interval, which may have a confidence level lower than desired. The pooling methods may need to be further evaluated in the future.
- Given the very small number of sets in which a non-zero bycatch was observed (positive sets), the panel recommended to use the grand mean of discard rates based on the pooled observed sets for all years and the annual logbook effort to produce annual discard estimates. Assuming the grand mean of discard rate based on all the pooled observed sets is a constant for the entire time series, and the trend of the discard estimates is solely driven by the logbook effort, which may need to be further evaluated in the future.
- The discard estimates and associated uncertainty estimates using the delta-lognormal method (SEDAR77-DW37 and SEDAR77-DW38) are regarded as an improvement over the discard estimates and associated uncertainty estimates using the ratio method reported in SEDAR77-DW20 and SEDAR77-DW21. More discard methods should be further explored in the future.

### *Indices*

1. During the assessment process, explore the utility of combining multiple indices into one scalloped hammerhead index using the Bayesian hierarchical model (Conn, 2009) or Dynamic Factor Analysis (Peterson et al., 2017). The data series that could potentially be combined as a recruitment index are Texas Parks and Wildlife gillnet series, Gulfspan gillnet series, South Carolina Coastspan Gillnet Long and Short Series and the Coastspan Longline Series.
2. Examine the utility of spatiotemporal modelling as a way to improve the indices of abundance for the NEFSC longline survey.

### *Ecological Factors*

- Improve understanding of all aspects of biology of hammerheads, particularly with regard to smooth and Carolina hammerhead occurrence, life history, and diet
- Investigate Bulls Bay, SC as a Habitat Area of Particular Concern for Carolina hammerhead

- Increase genetic surveillance to not only identify Carolina hammerhead individuals in the Atlantic, but also as a means to study use of nursery habitats and potential philopatry among all four species, potentially using close-kin mark-recapture techniques.
- Improve understanding of sex- and life stage-critical habitat for all species, particularly with regard to identification of essential habitat for data-poor species and life stages (Carolina and smooth hammerhead as well as young-of-year great hammerhead).
- Investigate impacts of environmental changes on life history characteristics, such as growth and reproduction
- Increase efforts in tagging and tracking to evaluate potential climate-induced range shifts
- Develop habitat suitability models for projecting climate-induced shifts in species distributions over time
- Increase effort for collecting environmental/oceanographic data with occurrence and movement data to identify linkages
- Assess the levels of environmental contaminants in hammerhead species and how those impact physiology and reproductive success
- Study the response of hammerhead species to harmful algal blooms and how those phenomena affect behavior and physiology

### **3. ASSESSMENT WORKSHOP RECOMMENDATIONS**

#### **Scalloped Hammerheads**

Support research to objectively rank (and prioritize) model sensitivities typically conducted during an Atlantic HMS domestic shark SEDAR stock assessment based on likely improvement for understanding model performance and providing robust management advice.

Support research to evaluate the effect of federal and state management actions, such as size restrictions and bag limits, on CPUE standardization and length composition of recreational catch available for use in stock assessment.

Support research to investigate trade-offs in model fit and uncertainty resulting from the use of selectivity functions with fewer parameters and informed priors.

Investigate the use of super years for length composition data with low sample size that result in poor quality annual length composition distributions.

Investigate the use of logistic selectivity vs dome-shaped selectivity for length composition data sets with the largest sizes. For example, asymptotic selectivity is typically implemented for fleets with the largest length size within an area as fleets approach including multiple fleets within a spatially-aggregated assessment model (e.g., HurtadoFerro et al. 2014; Punt et al. 2014).

However, this approach contrasts with evidence that asymptotic selectivity curves for length size are unlikely under equilibrium conditions (Waterhouse et al 2014).

Investigate the effect of reproductive output timing implemented within the Stock Synthesis for Atlantic HMS domestic shark stock assessment models on the resulting model fit and population dynamics.

### **Great Hammerheads**

- Since catches are dominated by recreational catches, decreasing the uncertainty associated with the recreational catches will be critical for improvement of future stock assessments of this stock.
- Since there are insufficient length composition data, programs to collect lengths to allow for a length-based, age-structured assessment in the future assessments should be developed.

### **Smooth Hammerheads**

Since catches are dominated by recreational catches, decreasing the uncertainty associated with the recreational catches will be critical for improvement of future stock assessments of this stock.

- Since representative length composition data can be added to SSS as they become available to free the input requirement of the stock status on the value of depletion in a given year (i.e. a proxy of overfished stock status), programs to collect length data to allow their incorporation into SSS in future assessments should be developed.
- Since there are insufficient time series data to develop indices of abundance, programs to collect relative abundance data to allow their incorporation into SSS in future assessments should also be developed.
- Since some of the life history data were borrowed from other stocks, programs to obtain representative biological information for this stock should also be developed.

## **1. REVIEW WORKSHOP RECOMMENDATIONS**

The Review Workshop was forced to end after one day due to a approaching hurricane, so the review process shifted to a desk review. As such, no Summary Report was produced, and therefore no research recommendations.